

Lane Perception for Automated Driving with Human-Anchored, Engine-Certified Ground Truth: An Explicit Counting Contract and Pre-Registered Single-Factor Evaluation

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Abstract

Lane perception for automated driving is bottlenecked by two problems that are usually conflated: the cost of pixel-accurate manual annotation, and the absence of an agreed definition of what a "lane line" is. We present a closed-loop research programme that attacks both. First, every label carries an explicit provenance credential on a three-level ladder -- human frame-by-frame revision, per-frame agent review, or engine-derived truth admitted through four certification gates -- so no claim has to guess what truth it rests on. The evaluation packs and the 75-frame training anchor are human-revised; the machine-generated increment that the ablations vary is engine-certified, and the rounds reported here added no new manual annotation. Second, we make the evaluation definition explicit and versioned: a *counting contract* gives every ratio an explicit numerator and denominator, and a *protocol* fixes the geometric scope of a line candidate (lateral extent, same-side merging, paint-appearance). We then evaluate one segmentation arm (six seeds) on two disjoint reference sets, on a road-disjoint one-shot held-out set, and in closed-loop driving. Three findings follow. (i) With the network weights held fixed, changing the post-processing switches and the evaluation scope changes the readings and can reverse the ranking of two arms; the two definitions each fail a *different* gate on the same data: under the paint-scope definition the adopted protocol clears all five gates on both sets (coverage 0.861, identity 0.716, precision 0.659, recall 0.885, role 0.794 on the limited-class pool), while the label-scope reading of the same checkpoints fails recall (0.559 on the development pool) because about 30% of annotated line pixels do not satisfy the appearance predicate the adopted definition uses (their visual identifiability is not independently verified in this work). (ii) The one-shot held-out confirmation does **not** reproduce those numbers (label-scope recall 0.460, precision 0.763), and we report that as the headline result of the independent test rather than hiding it. (iii) In closed-loop driving the same definition change was accompanied by no observed centre-line crossing in four runs (baseline: 8, 8, 8 and 13 frames) and by a collapse of availability (sensor lane-source rate median $0.851 \rightarrow 0.292$, a 66% relative drop), with the car barely moving; the driving acceptance gate still fails 0/4 runs, and we localise the remaining blocker with measurements rather than conjecture: a static deadlock in which a stationary, off-centre car has its generated path refused by a 4 m body-sweep rule. Throughout, changes are pre-registered, tested one factor at a time, and judged by rules fixed before the runs; negative results are reported with the same detail as positive ones. All figures and verdict files are regenerated from the recorded run artefacts.

Index Terms—Lane detection, ground-truth generation, evaluation protocol, automated driving,

Highlights

- Ground-truth provenance is credentialed and enforced: human revision, agent review, engine-certified.
- A counting contract gives every reported ratio an explicit numerator and denominator.
- The protocol hash binds a sealed held-out set to the definition it is read under.
- The two natural definitions of a lane line fail different gates on identical data.
- A one-shot held-out confirmation does not reproduce the development-set numbers.
- Five pre-registered driving factors were rejected; the blocker is a measured deadlock.

I. Introduction

Lane-keeping is the lowest-level competence of an automated driving stack, and it is still surprisingly hard to evaluate. Two obstacles recur in the literature. The first is annotation cost: pixel-accurate lane labels are expensive, so public datasets are small relative to the diversity of road appearance, and every new operational domain demands a new labelling campaign. The second is definitional: papers report "lane detection" accuracy against ground truth that may include or exclude faded, occluded, dashed, yellow, or shadowed markings, and may or may not count markings beyond the ego lane. Two systems can therefore disagree by tens of accuracy points while disagreeing about nothing except the definition of the object they detect.

This paper reports a research programme built around those two obstacles. It is not a new network architecture; it is an attempt to make the *data* and

the *evaluation* rigorous enough that architectural progress can be measured. The programme has three components.

Credentialed truth provenance. Every label in this programme carries a provenance credential, and the credential is enforced rather than declared: a command-line claim cannot promote a label (Section III-B). The ladder has three rungs -- human frame-by-frame revision, per-frame agent review, and engine-derived truth -- and the engine rung is admitted only through four explicit certification gates: a credential gate (the label source must be declared and verifiable), a paint-appearance gate (the pixels claimed as line must look like paint in the image), a road-type gate (the frame must carry the road-type column that makes pavement decidable), and a negative-evidence gate (frames claimed to contain no line must show evidence of their own, not merely the absence of a label). The rounds reported here add no new manual annotation: the experimental variable is the composition of the engine-certified increment on top of the human-revised anchor, and Section VII states what the anchor's small size costs us.

An explicit counting contract. Every reported ratio is defined by a counting contract that names its numerator and denominator over an explicit set chain: frames with candidates, extracted candidate instances, reference instances present in the annotation, matched instances, instances whose lateral placement agrees, and instances whose left/right role agrees. Coverage, identity and role are therefore instance-level ratios with stated denominators; precision and recall remain pixel-level and are reported in two *scopes* (label pixels, and paint-like pixels) because the two answer different questions.

Versioned protocol definitions with hash-gated evaluation. A protocol fixes the geometric scope of a line candidate and is a first-class, versioned artefact. A protocol change alters the evaluation hash, and both the frozen held-out set and the one-shot confirmation record that hash, so a confirmation produced under one definition cannot be silently attributed to

another. This mechanism caught a real defect during the work: before it was fixed, the hash did not follow the protocol, and a set sealed under the old definition would have authorised a confirmation under the new one.

Method discipline. Every change in this paper was pre-registered with its reading rule before the runs; the rules include safety non-inferiority read as a maximum over runs, an explicit refusal to treat an unmeasured quantity as a pass, and a prohibition on relaxing the frozen gates. Changes were applied one factor at a time and judged against alternating control/factor runs. Where a pre-registered rule produced a verdict we considered too coarse, we report the verdict as given and record the rule's defect rather than reinterpreting the result; one such defect — safety metrics that are frame *counts* and therefore scale with how far the car travelled — is analysed in Section V-F and quantified in 26.

Contributions. (1) A credentialed truth-provenance ladder (human revision / agent review / engine-certified) with enforcement, plus a certification-gated engine pipeline for the machine-generated increment; the composition of every pool is stated explicitly, and the reported rounds add no new manual annotation. (2) The counting contract, and the demonstration that the two natural readings of "line" fail different gates on the same data — a definitional disagreement that moves about 30% of annotated line pixels out of the adopted scope (which retains 69% of annotated line pixels on the development pool and 89% on the limited-class pool). (3) A protocol/version/hash mechanism that binds a held-out set to the definition under which it is read. (4) A one-shot, road-disjoint confirmation with an explicit UNKNOWN policy, whose negative result (no reproduction of the development-set numbers) we report as the finding. (5) A measured localisation of why a perception stack that passes its offline gates still fails a closed-loop acceptance gate, including a static-deadlock mechanism and the observation that the "on-pavement" gate was in fact being driven by a neural obstacle layer rather than by pavement evidence.

II. Related Work

Lane detection methods. Modern lane detection is dominated by deep segmentation and by parametric or anchor-based curve regression. Encoder-decoder segmentation with dilated convolutions and a discriminative loss (LaneNet [1]) and message-passing across rows (SCNN [2]) established the segmentation line of work that the present system follows; cross-layer refinement with anchor-based heads (CLRNet [3]) improved curve accuracy and speed. Chinese-language work in the same family explores lightweight variants of CLRNet [4], B-spline end-to-end regression [5], graph/transformer hybrids [6], transformer with cross-attention [7], parallel detection over image sequences [8], multi-scale feature aggregation [9], DeepLabV3+ variants [10], symmetry-prior and key-point methods for complex scenes [11], and 3-D lane estimation with convolutional architectures [12]. These works share our object of study but differ in what they optimise: they report accuracy against datasets whose annotation conventions differ from each other, which is exactly the ambiguity this paper makes explicit.

Datasets and ground truth. TuSimple [13] and CULane [2] remain the standard benchmarks; their annotation conventions (lane presence per row, presence of far-field markings, treatment of occlusion) differ, and neither reports the annotation's *appearance* properties. Ground-truth generation has been studied as a problem in its own right: Borkar et al. [14] built an automated ground-truth process around time-slicing visualisation for a night-time lane detector, and McCall and Trivedi [15] surveyed and evaluated video-based lane estimation with an explicit eye on evaluation methodology. Our work continues that line with two additions: truth derived from engine state rather than from imagery, and a counting contract that makes the denominators of every reported ratio inspectable.

Automatic labelling and simulation. Self-training and auto-labelling are widely used to reduce annotation cost, but their evaluation inherits the definitional problem above. Simulator-derived labels are common in synthetic benchmarks; less common is the discipline of certifying each generated label against image evidence, which we adopt here because a label that cannot be recovered from the image is not a learnable target.

Evaluation discipline. Pre-registration is standard in empirical sciences and rare in perception research. Our contribution here is not the concept but a worked, fully recorded instance: a pre-registration, an alternating single-factor A/B, a fixed reading rule, and a verdict per factor — including five factors whose verdict was negative.

III. System and Method

A. Overall architecture

The stack is a perception-led driving system. A camera ring (eight mounts) and a LiDAR range image feed a semantic segmentation head whose line output is converted to world-space markings; a lane reference is selected from those markings under the protocol definition; a planner produces a path; a safety monitor arbitrates the tick against hard checks (pose, boundaries, occupancy, obstacle risk) and publishes a speed and level; the controllers consume the arbitrated path. The iron rule of the stack is that the lateral reference comes from perception only — the map answers "where to", never "where the lane is". 1 shows the candidate-scope pipeline with the measured revocation rate discussed in Section V-E.

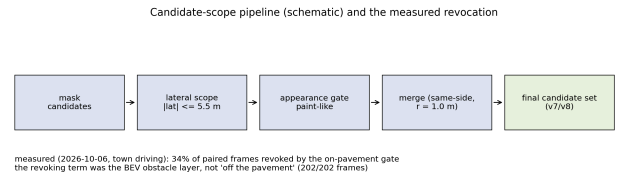


Fig. 1. Candidate-scope pipeline (schematic) with the measured revocation rate.

B. Truth provenance, credentials, and certification gates

The programme distinguishes three rungs of truth and records the rung with every label: **human revision** (a human annotates or corrects the line labels frame by frame), **agent revision** (a per-frame agent review of engine output), and **engine-certified** truth (the engine's own line geometry projected into the camera). Two pools are human-revised: the training anchor (5 directories, 75 frames) and the development/evaluation packs (9 packages, 117 frames), which is why the acceptance results of Section V-A are measured against human annotation. The machine-generated increment that the ablations vary is engine-certified; it is admitted only when: (i) the credential is present and parseable, and the label source is declared (frames whose provenance cannot be named are excluded rather than guessed); (ii) the pixels claimed as line satisfy a paint-appearance predicate — white paint requires high value, low chroma and a red/blue balance; yellow paint requires a hue ratio band; the predicate is a single-source module so that a threshold change is a versioned change, not a local edit; (iii) the frame carries the road-type column, without which "on pavement" is undecidable; (iv) a frame claimed to contain no line carries positive evidence of its own (a certified negative), so that "no label" is never read as "no line". Frames failing any gate are excluded from both training and evaluation and are counted in the audit, not silently dropped.

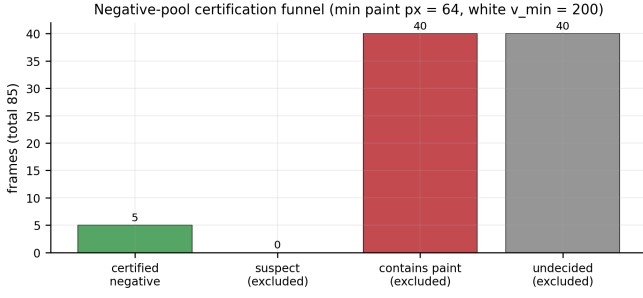


Fig. 2. Negative-pool certification funnel: frames admitted as certified negatives versus excluded categories.

The gate bookkeeping is itself a result: 2 shows the negative-pool funnel for one collection round (admitted, suspect, contains paint, undecided) — a pool that looked plentiful on disk shrank to a certified subset before it was ever used as training signal.

C. The counting contract

The contract is stated over integer counters that are accumulated per frame; the ratios are computed once from those integers, never by averaging per-directory ratios. P counts frames whose annotation carries a line on pavement (decided frame by frame: a frame is not admitted because another frame in its group has a line); C counts the perception candidates inside P frames that enter the evaluation; C_{out} counts candidates on frames outside P and is reported separately; $R \subseteq C$ counts candidates whose own side has an available reference; $M \subseteq R$ those satisfying the frozen matching condition; $L \subseteq M$ those where both the prediction and the reference have a decidable left/right role; and $A \subseteq L$ those whose role agrees. The reported instance-level ratios are coverage $|R|/|C|$, identity $|M|/|R|$ and role agreement $|A|/|L|$; pixel-level precision and recall are reported in two scopes (label, paint). 3 shows the contract.

Two consequences have to be read off this definition rather than assumed. First, coverage is the **candidate-reference availability**: the share of evaluated candidates whose own side has truth to compare against, not the share of true markings that were detected. A line-bearing frame on which the perception produces no candidate contributes nothing to $|R|/|C|$; its miss is carried by pixel-level recall instead. Second, candidates on frames outside P are counted as C_{out} and reported, so that "no reference available" is never scored as a coverage failure. We therefore claim no instance-level recall: the three conditional ratios above cannot substitute for one, and reporting one would require the total number of ground-truth instances together with an explicit one-to-one assignment rule. Every ratio still has an explicit denominator, and the frame-level counter P is reported separately from the candidate-level counters, so a run with few candidate-bearing frames cannot masquerade as a good one.

Counting contract v5 (schematic): every ratio has an explicit numerator and denominator

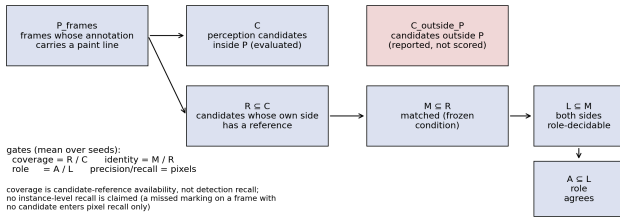


Fig. 3. Counting contract (schematic): every ratio has an explicit numerator and denominator, over the frozen counter definitions; coverage is candidate-reference availability, not detection recall.

D. Protocol definitions

A protocol fixes three things about a line candidate: its **lateral scope** (candidates farther than a threshold from the ego's own lane are not "my lane's lines"), whether **same-side near-duplicate markings are merged** into one instance, and whether an **appearance gate** removes candidates whose pixels do not look like paint. The frozen baseline protocol applies none of the three (lateral scope off, merging off, appearance off); the adopted protocol applies all three with a lateral threshold of 5.5 m, a merge radius of 1.0 m and the appearance predicate of Section III-B. 5 shows the switch matrix and the live hash; the hash is a function of the active protocol, so evaluation artefacts cannot be reattributed across definitions.

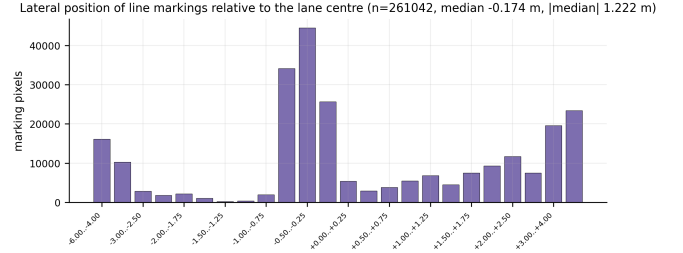


Fig. 4. Lateral position of line markings relative to the lane centre, pooled over the human-revised development pool.

The lateral scope is not arbitrary: it is set from the measured geometry of the reference pool, shown in 4 (261 042 marking pixels, median offset -0.17 m, median absolute offset 1.22 m). A scope of 5.5 m retains the own lane's markings and their immediate neighbours while excluding the far-field markings that dominate the false pairings discussed in Section V-E.

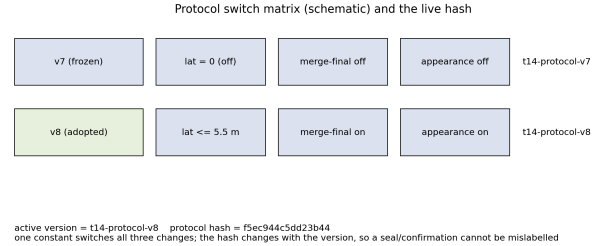


Fig. 5. Protocol switch matrix (schematic) and the live evaluation hash.

E. Training-composition design space

Because ground truth is generated rather than annotated, the composition of the training pool is a first-class experimental variable. We study it along one axis: the **negative dose**, i.e. how many certified line-free frames are mixed into the pool relative to the base composition, together with composition variants that place markings in specific geometric relations (near/far pairs, same-side pairs). Section V-C reports the resulting dose-response.

F. Held-out final set: sealing, hash gating, one-shot consumption

An independent confirmation set is composed under three constraints: it must be **road-disjoint** from every training pool (identity taken from the generation metadata, not from directory names), it must be certified by the gates of Section III-B, and its content must be frozen by a digest over per-frame hashes. Sealing records the frame digests, the protocol hash and the composition; access is then gated on the protocol hash, and a single confirmation **consumes** the set: after a successful confirmation the set may not be read again, and a refused access is recorded in a ledger without consuming it. The confirmation program evaluates the candidate on the sealed frames and records which quantities were measured; quantities that

were not measured are recorded as UNKNOWN and may not be reported as confirmed. The full flow, with the live seal, is shown in 6.

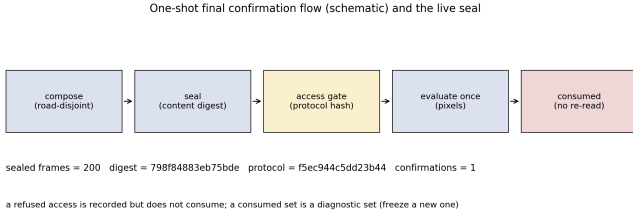


Fig. 6. One-shot final-confirmation flow (schematic) with the live seal.

G. Driving stack and safety arbitration

In closed loop the same perception feeds a planner and a safety monitor whose arbitration ladder evaluates rules in a fixed order; soft rules accumulate speed caps, while a rule whose worst level is `minimal_risk` stops the tick. 7 lists the ladder. Two rules matter for Section V-E: the **planned body-sweep rule**, which stops the tick when the body sweep of the planned path would cross a lane boundary closer than a hard threshold (4 m), and the **on-pavement gate**, which revokes the perception lane when the lane centre does not lie on observed drivable surface.

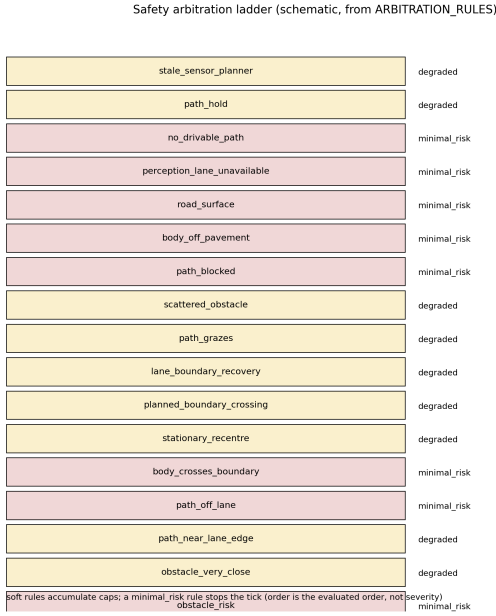


Fig. 7. Safety arbitration ladder (schematic, from the monitor's rule registry).

IV. Experimental Setup

Reference sets. Two disjoint sets are used for the primary acceptance: a development pool of 9 certified scenes (117 frames) and a limited-class pool of certified scenes chosen for near-field background structure (24 frames per arm). Neither shares frames with training pools.

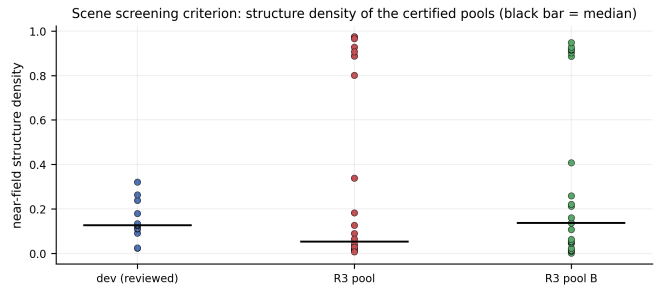


Fig. 8. Scene screening criterion: near-field structure density of the certified pools (black bar = median).

The limited-class pool is selected, not sampled: candidate scenes are ranked by a model-independent near-field structure-density criterion computed from the human annotation (8), so the pool concentrates on the scenes where identity and role are hardest rather than on the average scene. A third, road-disjoint set of 200 frames (96 line-bearing) is sealed for the one-shot confirmation of Section V-D.

Disjointness is audited with pose, not with names: every candidate collection is checked against the development anchors under a 50 m buffer and gets a recorded verdict (supplement). The same machinery rejected the first two compositions of the sealed set, which is why the sealed composition also carries a content-level audit (19).

Arms. All segmentation arms share one initialisation, one budget and one recipe; they differ only in training-pool composition (the dose arms) or in a single post-processing factor (protocol components). Composition is explicit: a human-revised anchor of 5 directories (75 frames) plus a varying number of engine-certified packages (2 paired-line packages of 16 frames and 12 certified negative packages of 96 frames in the reported base), with the negative dose as the varied axis. Six seeds are used for acceptance; composition arms are reported with their seed counts.

Metrics and gates. The frozen gates are coverage ≥ 0.80 , identity ≥ 0.60 , precision ≥ 0.40 , recall ≥ 0.70 and role agreement ≥ 0.70 ; recall is read in the paint scope for the adopted protocol and in the label scope for the baseline, and both readings are always reported. The verdict rule changed between the two rounds and the change is part of the record rather than a detail: the baseline round's stored verdicts require **every seed** to pass, while the adopted round stores `verdict_mode = mean` and reports the per-seed values alongside. Section V-A therefore gives the stored verdicts and a re-tabulation of the same per-seed numbers under each rule, so that no gate outcome has to be read under an unstated rule.

Driving experiments. Closed-loop runs use one fixed scenario (a town link), alternating control and factor arms, four runs per arm; hard targets are the benchmark's fixed checks (frames, reversing, centre and edge crossings by reference line *and* by body sweep, stall, on-road, collisions, goal). Reading rules were fixed before each round: hard gates require 4/4, safety items are compared as the maximum over runs, availability as median with range, and an UNKNOWN never releases a gate.

Discipline. Every factor in Section V was pre-registered with its reading rule; no threshold was relaxed at any point;

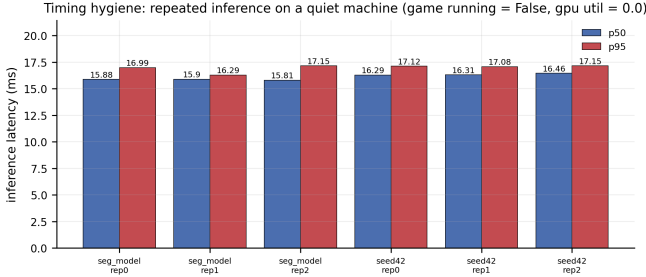


Fig. 9. Timing hygiene: repeated inference measurements on a quiet machine.

Timing hygiene is recorded rather than estimated: 9 is the repeated-measurement protocol used before any latency claim (quiet-machine precondition, repeated runs, smaller value taken — a leaked simulator instance once inflated p95 by a factor of 7.6). every verdict file, manifest and per-frame telemetry trace is retained, and every figure in this paper is regenerated from those files by two scripts.

V. Results

A. Acceptance under two definitions

Table I reports the two definitions on both sets, and Table II re-tabulates the recall gate of the same checkpoints under both post-processing settings and both scopes. Under the baseline definition the stored verdict fails identity on **both** sets (0.433 dev, 0.414 limited-class) and additionally precision (0.402) and role (0.674) on the limited-class set; under the adopted definition all five gates pass on both sets by the mean rule (coverage 0.956/0.861, identity 0.665/0.716, precision 0.894/0.659, paint recall 0.812/0.885, role 0.883/0.794). 11 shows the same numbers against the thresholds.

Three qualifications belong with this table. (i) The two rows were not judged by the same rule. Re-tabulated under a common mean rule, the

Set	Definition	Coverage	Identity	Precision	Recall	Role
Dev (9 scenes, 117 frames)	baseline (label scope)	0.952	0.433	0.610	0.736	0.868
Dev	adopted (paint scope)	0.956	0.665	0.894	0.812	0.883
Limited-class (24 frames)	baseline (label scope)	0.819	0.414	0.402	0.804	0.674
Limited-class	adopted (paint scope)	0.861	0.716	0.659	0.885	0.794

Table II. The recall gate re-tabulated: the same six checkpoints per pool, read under both post-processing settings and both scopes, with the number of seeds below the

Pool	Post-processing	Scope	Mean recall	Seeds below 0.70
Development	baseline switches (v7)	label	0.736	2/6
Development	baseline switches (v7)	paint	0.813	1/6
Development	adopted switches (v8)	label	0.559	6/6
Development	adopted switches (v8)	paint	0.812	1/6
Limited-class	baseline switches (v7)	label	0.804	0/6
Limited-class	baseline switches (v7)	paint	0.891	0/6
Limited-class	adopted switches (v8)	label	0.775	0/6
Limited-class	adopted switches (v8)	paint	0.885	0/6

baseline's limited-class precision (0.4018) is above its 0.40 threshold even though 4 of 6 seeds fall below it, and the baseline's development recall (0.736) is above its 0.70 threshold even though 2 of 6 seeds fall below it; under a common all-seeds rule the adopted definition's development label-scope recall fails 6 of 6 seeds while its paint-scope recall fails 1 of 6. (ii) The protocol change moves three post-processing switches and the evaluation scope at the same time, so this table cannot attribute the whole difference to the definition; Table II separates the scope from the post-processing for recall, and the per-switch single-factor results are in Section V-B. (iii) The candidate-level gates are the stable part of the picture: the identity gate is missed by 6 of 6 baseline seeds on both sets and role by 5 of 6 on the limited-class set, while under the adopted definition no seed misses any of the three (10).

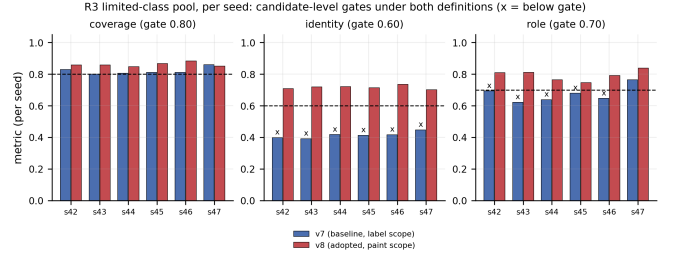


Fig. 10. R3 limited-class pool, per seed: the three candidate-level gates under both definitions, drawn from the same verdict files as Table I (x = below gate).

10 plots the limited-class pool seed by seed under both definitions, with each candidate-level gate marked when missed; it is drawn from the same two verdict files as Table I, so the figure and the table cannot disagree about which run produced them.

Table I. The two protocol definitions on both reference sets, as stored by each round under its own verdict rule (means over six seeds; bold marks a gate the stored verdict failed; the adopted protocol is read in paint scope, the baseline in label scope). The rule differs between the two rows — see the text and Table II.

0.70 threshold. The paint scope reads each checkpoint under the adopted definition; the label scope under the baseline one.

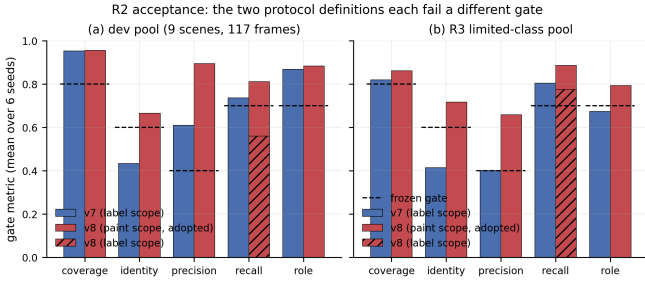


Fig. 11. Gate metrics against frozen thresholds for the two protocol definitions, development pool (a) and limited-class pool (b).

B. The disagreement is definitional

Two things move together here and they have to be separated. The **scope** changes the reading: Table II shows that under a fixed post-processing setting, moving from the label scope to the paint scope changes development recall from 0.736 to 0.813 (baseline switches) and from 0.559 to 0.812 (adopted switches) — in the second case a 25-point swing with no change to the weights. The **post-processing** also changes the reading, and not in the same direction: with the scope held at label, the adopted switches cost 18 points of recall ($0.736 \rightarrow 0.559$), because the lateral scope, the same-side merge and the appearance gate all remove candidates that the label scope counts. What the scope change is worth in pixels is measured, not assumed: the paint scope retains 69% of annotated line pixels on the development pool and 89% on the limited-class pool, and 13 shows the per-seed label-scope and paint-scope recall with the non-paint fraction of label line pixels overlaid (~ 0.30). Those excluded pixels are the ones the appearance predicate does not select; Section VI states what that does and does not license us to claim.

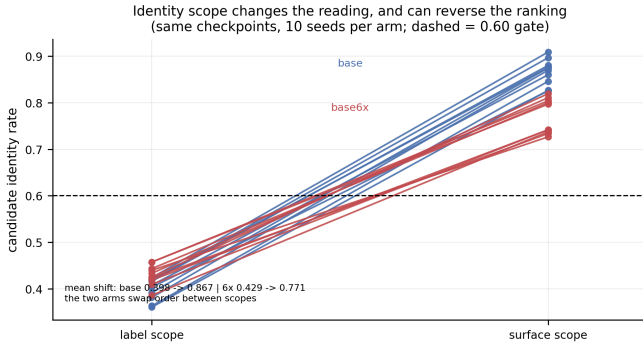


Fig. 12. Identity scope changes the reading and can reverse the ranking: 20 checkpoints (two arms, ten seeds each) read in label and surface scope.

The instance-level gate is more sensitive than that, and in a way that matters for model selection. 12 reads 20 checkpoints (two arms, ten seeds each) under the label-scope and surface-scope identity definitions: the base arm moves from a mean of 0.399 to 0.867 and the $6\times$ arm from 0.429 to 0.771. The absolute values nearly double for one arm and fall for the other, and **the ranking of the two arms reverses** (base below $6\times$ by 0.031 under the label scope, above it by 0.095 under the surface scope). The same reversal is recorded in the repository's protocol proposal for the surface scope, which is why that scope is reported rather than promoted: a gate that can change which checkpoint is selected is a model-selection decision, not a reporting detail. An identity threshold must therefore name its scope, exactly as the recall gate does. This experiment uses its own pool and its own ten seeds per arm, so its numbers are not comparable with Table I's six-seed means and are not presented as such.

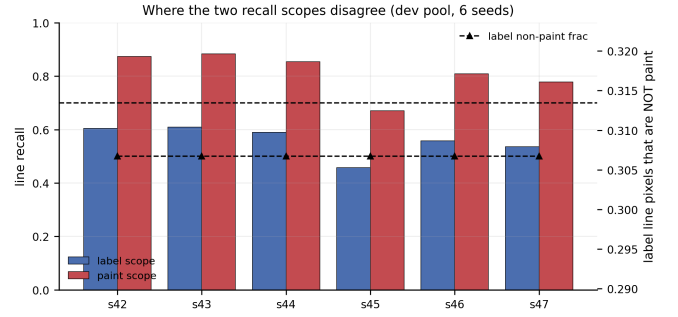


Fig. 13. Per-seed label-scope versus paint-scope recall with the non-paint fraction of label line pixels (development pool).

Two further single-factor results constrain how the adopted definition behaves. 14 (boundary map) sweeps mask dilation: dilation buys recall in both sets but costs precision, and the limited-class set pays much more for it (at one pixel of dilation its precision falls from 0.65 to 0.39, a drop of 0.26, against 0.89 to 0.84, a drop of 0.05, on the development pool; the two rows are read at their own dilation settings), so no single dilation value clears both sets. 15 shows the appearance gate acting exactly as designed: precision rises sharply (dev $0.650 \rightarrow 0.884$; limited-class $0.387 \rightarrow 0.650$) and IoU improves on the limited-class set ($0.347 \rightarrow 0.532$) at the cost of recall (dev $0.766 \rightarrow 0.595$).

The appearance predicate is a mask gate, not a truth criterion: its own module documents 13k–68k px of hits on certified line-free frames (bright pavement and bright gravel both satisfy the white branch), which is why it is used to remove candidates rather than to declare where lines are. With that caveat, the instance-level probe explains why it can afford to be aggressive: across the certified limited-class scenes the reference instances are covered by annotation at essentially 1.0 regardless of lateral offset, so a surviving candidate is still measured against complete truth. The other post-processing lever, the parallel-merge radius, removes near-duplicate same-side candidates (dev 477 $\rightarrow$ 166 kept as the radius tightens from 6.0 m to 3.0 m) at a cost in identity, which is why the adopted radius is 1.0 m rather than the aggressive end of the sweep. The two mask thresholds were swept as well; on this pool the negative side is not measurable (no eligible negative frames), so that sweep is reported in the supplement rather than plotted as a zero.

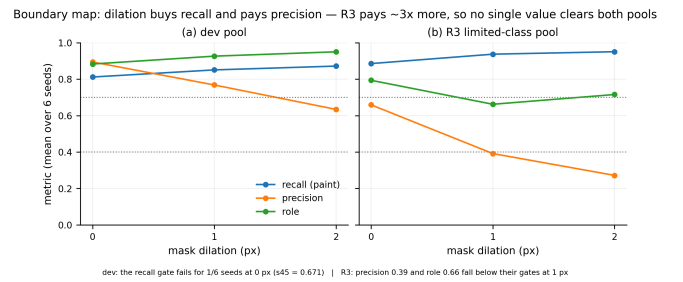


Fig. 14. Boundary map: dilation buys recall and pays precision; the limited-class set pays about three times more.

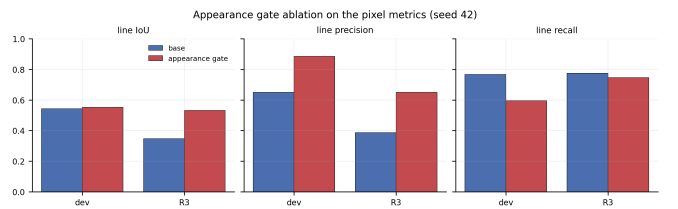


Fig. 15. Appearance-gate ablation on the pixel metrics (line IoU, precision, recall) for both sets, seed 42.

C. Training-composition dose response

16 shows the identity rate against the negative dose, per seed and as a mean: 0.707 (base composition, one seed), 0.711 (4 $\times$, six seeds, 0.667–0.732), **0.763 (6 $\times$, one seed)** and 0.701 (8.5 $\times$, six seeds, 0.655–0.744). The seed counts are unequal by construction (the 6 $\times$ point is the delivered candidate's composition and rests on the seed that was delivered), so the response supports a narrower claim than a dose–response curve usually carries: the highest *observed* reading sits at 6 $\times$, above the 4 $\times$ and 8.5 $\times$ means and above every individual 4 $\times$ /8.5 $\times$ seed, but a single seed cannot establish a stable optimum. The comparison that would settle it — the same doses with paired seeds and equal counts — is listed as unfinished work, and the 6 $\times$ arm remains the delivered composition because it was selected before the confirmation, not because the curve proves it optimal.

The programme's training history (321 recorded runs across 44 families, supplement) converges within 20–40 epochs of the 120-epoch budget, and families whose line channel was masked sit at zero line IoU by construction rather than by failure (annotated in the figure), so their curves are not negative results. 17 sweeps the lateral scope instead: tightening it from 6.0 m to 3.0 m raises identity (dev 0.582 $\rightarrow$ 0.713; limited-class 0.581 $\rightarrow$ 0.758) and removes candidates (477 $\rightarrow$ 166 and 166 $\rightarrow$ 127), while role agreement is flat — the lateral scope is an identity lever, not a role lever. Two further single factors were rejected: stochastic weight averaging over the last five epochs rescued no seed and a Tversky loss-weight arm multiplied the false-positive frame rate on the negative side by 4.5 (11.5% $\rightarrow$ 51.9% of eligible frames, +40.4 percentage points; 329 $\rightarrow$ 1730 false-positive pixels; maximum component 320 $\rightarrow$ 808 px), which is why the delivered arm keeps the balanced loss. Across every recorded pixel arm the negative-side readings cluster rather than spread (supplement), which is why the negative-example dose had to be varied instead of a loss-weight knob. Extending the dose response to five metrics (supplement) shows that identity and role move with the dose while coverage and the off-road candidate fraction barely do. The availability cost of the adopted definition is also visible at the pairing stage itself: 3–4 paired frames lost per 117-frame run.

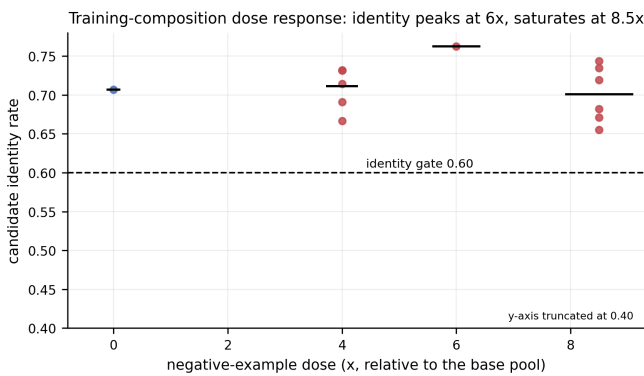


Fig. 16. Dose response of the candidate identity rate (per seed and mean).

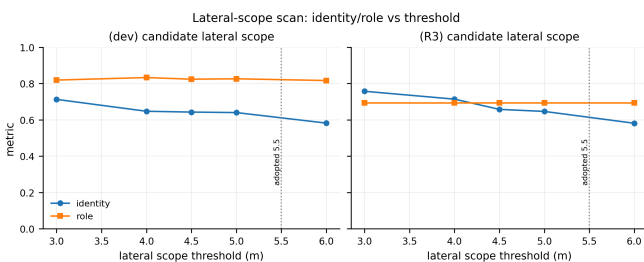


Fig. 17. Lateral-scope scan on both sets: identity and role versus threshold, with the number of retained candidates.

D. One-shot held-out confirmation

The sealed set was read once, by the delivered candidate. Overall label-scope recall is **0.460** and precision **0.763** (IoU 0.403; road IoU 0.633); per group, three of seven groups sit at recall 0.17–0.20 — but not with the same precision: two of them hold precision 0.864 and 0.922 while the third (a27) has precision **0.169** and sends 61% of its predicted line pixels off the road, so it is a group the candidate largely fails rather than a clean high-precision group. One group carries no line ground truth (recall undefined, recorded as such), and the remaining groups range 0.43–0.63. On the negative side 22 of 105 eligible frames carry false line pixels (20.9%), but those pixels are 0.084% of eligible pixels, the largest connected component is 1164 px, and no false candidate appears inside the control region. 20 shows the confirmation. Two properties of this result deserve emphasis. First, it does **not** reproduce the acceptance numbers of Section V-A: the independent set is harder and its group composition differs. Composition is counted two ways and both are recorded: by package, 104 of the 200 frames come from certified line-free packages and 96 from line-bearing ones; by per-frame eligibility, 95 frames carry line truth and 105 are eligible negatives, the difference being frames whose own eligibility differs from their package's label. The delivered candidate's claim is bounded by the numbers reported here rather than by the acceptance table. Second, the confirmation records that the instance-level probes (identity, role, reference coverage) were **not** run on this set; they are UNKNOWN and are not reported as confirmed.

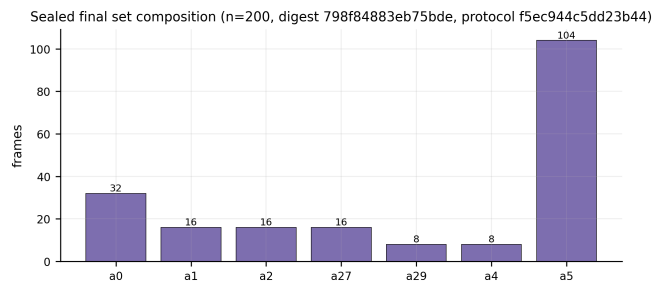


Fig. 18. Sealed final-set composition by group, with digest and protocol hash.

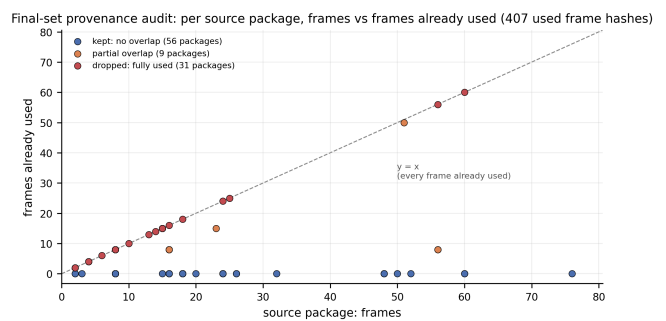


Fig. 19. Final-set provenance audit: content-level overlap of each source package with the used frame set.

18 states the composition the confirmation ran on (200 frames across seven groups; digest and protocol hash printed in the title so the figure cannot be reattributed to another seal). 19 is the audit that admitted it: every source package is compared against the 407 frame hashes already used, and the packages that showed overlap were dropped before sealing. Third, and relevant to the gap with Section V-A, the sealed set's truth is **engine-certified** while the acceptance sets are **human-revised**: the two numbers are measured against different truth rungs, not only different data.

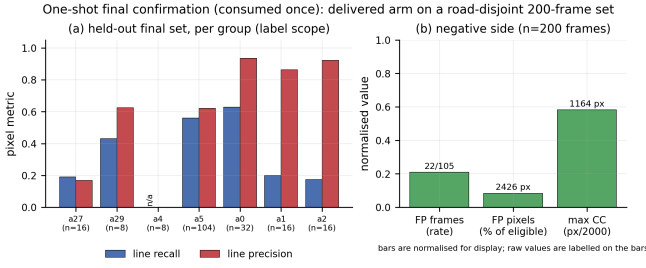


Fig. 20. One-shot final confirmation on the sealed road-disjoint set: per-group metrics (a) and negative-side diagnostics (b).

E. Closed loop

Definition change in closed loop. With the baseline definition the closed-loop runs show centre crossings of 8, 8, 8 and 13 frames; with the adopted definition no centre crossing was observed in any of the four runs. That is a statement about four runs, not a rate: the same runs travel very little (stall fraction 0.93–1.00, see below), and zero events at near-zero exposure cannot be extrapolated. The adopted definition pays for it with availability: the sensor lane-source rate falls from a median of 0.851 (range 0.231–0.975) to 0.292 (0.097–0.524), a 66% relative drop, and the paired-lane rate from 0.890 (0.809–0.986) to 0.528 (0.130–0.995). 21 shows the trade-off. Both effects are consistent with the offline finding that the baseline keeps far and non-paint candidates whose pairings pull the car across the lane centre.



Fig. 21. Closed-loop trade-off: centre crossings versus sensor lane-source rate, four runs per arm.

Acceptance failure and its mechanism. The driving acceptance gate (goal + perception-led lateral mode + strict arbitration, four runs) fails 0/4: stall fraction 0.93–1.00, travelled 0.9–14.5 m in 120 s, 76.5–88.8 m remaining to goal; the safety-side checks that were measured held (0 collisions, 0 off-road frames, 0 reversing). 22 shows the per-run hard-target matrix over the recent runs. We then localised the blocker by measurement rather than conjecture. Over 12 town runs, the frames whose reason is "planned vehicle body crosses lane boundary" number 512; in those frames the car is stationary in 96% of cases, the current body is inside the lane in 512/512, and the planned crossing sits at a median of 2.50 m (509/512 within 2–4 m) — i.e. just inside the 4 m hard-stop threshold. The car is parked 0.86 m off the lane centre with its heading $\sim 11^\circ$ off the lane bearing, so the inflated body sweep of the path the planner actually generated crosses the boundary within ~ 2.5 m. What the record supports is that the *generated* path is refused, repeatedly, in every frame of that set; it does not establish that no dynamically feasible forward path exists, because the planner's candidate set is not the set of all feasible paths. A controlled sweep over fixed poses is listed as unfinished work. 23 shows the anatomy.

Per-run traces (speed and path-to-lane deviation), arm-level distributions and the lane-placement distributions are in the supplement; the safety margins measured in the same runs (closest obstacle, minimum time-to-collision, path occupancy) are what backs the statement that the acceptance failures are availability failures, not safety failures.

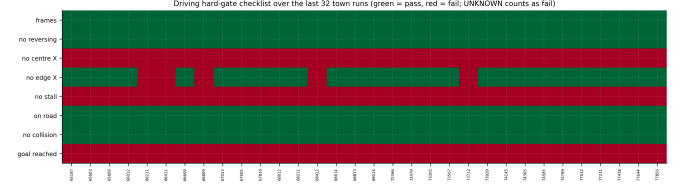


Fig. 22. Driving hard-target checklist over the most recent town runs (green = pass, red = fail; an unmeasured target counts as fail).

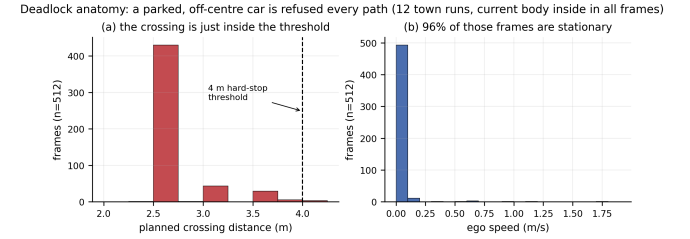


Fig. 23. Deadlock anatomy: the planned crossing sits just inside the 4 m threshold (a) and the car is stationary in 96% of those frames (b).

Four factors run, one proposal refuted before implementation. We pre-registered five driving-layer factors against this blocker; four were implemented and run, and the fifth (pavement re-centring) was refuted by prior measurement before any code was written (Table III). A bounded "hold" window that lets a verified trajectory be reused past its offer age raised motion (reuse 7 $\rightarrow$ 42 frames; travelled 3.15 $\rightarrow$ 8.5 m) but worsened the worst-case body-crossing count (29 $\rightarrow$ 30). Pulling the planned path onto the perceived lane centre engaged in 39 frames and improved the worst case substantially (centre crossings 3 $\rightarrow$ 0; body crossings 84 $\rightarrow$ 39) but worsened one right-side item (0 $\rightarrow$ 2). Removing the obstacle term from the on-pavement gate — see below — doubled travelled distance (6.85 $\rightarrow$ 14.1 m) and halved the distance to goal (83.5 $\rightarrow$ 76.6 m) but worsened body crossings (23 $\rightarrow$ 66) while the car moved twice as far. Swapping in the delivered perception arm produced no separation on path availability (0.179 $\rightarrow$ 0.200, overlapping ranges) and a worse worst case (36 $\rightarrow$ 49). All four implemented factors follow the pre-registered maximum rule and are reported as negative; none was adopted, and the fifth remains untested by construction.

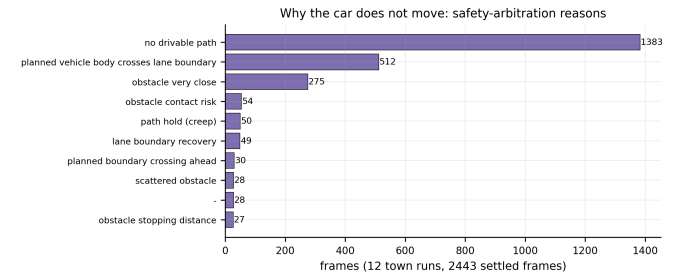


Fig. 24. Why the car does not move: safety-arbitration reasons over 2443 settled frames.

The arbitration histogram is the evidence behind the stall fraction: over 2443 settled frames (frames at least 8 s into a run, pooled over the twelve town runs the figure names), "no drivable path" is the largest single reason at **1383 frames**, followed by the planned body-sweep crossing at **512**, which together account for the bulk of the stationary time.

Table III. The pre-registered driving-layer factors against the deadlock, with mechanism engagement, motion read-outs and the pre-registered verdict. Four were

Factor	Mechanism engaged?	Travelled (median)	Stall (median)	Worst-case body-cross frames (control → factor)	Verdict
Bounded hold window	yes (reuse 7 → 42 frames)	3.15 → 8.5 m	0.986 → 0.958	29 → 30	rejected
Path re-centring	yes (39 frames)	5.1 → 5.75 m	0.984 → 0.980	84 → 39	rejected (one item worse)
On-pavement gate (obstacle term removed)	yes (availability 0.07–0.52 → 0.36–0.90)	6.85 → 14.1 m	0.966 → 0.950	23 → 66	rejected
Delivered perception arm	no (no separation)	—	0.979 → 0.995	36 → 49	rejected
Pavement re-centring (as proposed)	not implemented	—	—	—	refuted by prior measurement

Why the lane was being dropped. The on-pavement gate revokes the perception lane when the lane centre does not lie on observed drivable surface. Instrumenting the gate's evidence over two diagnostic runs (415 settled frames) shows that of 202 revoked frames, **all 202** were revoked by the obstacle term (median 6 obstacle cells of ~12 observed samples), while the number of samples that were observed but outside the drivable mask is zero in 192 of them. The obstacle layer is populated from a neural bird's-eye-view head, so on this link the "on-pavement" gate was in effect an obstacle-prediction veto on the lane centre. 25 shows the evidence and the availability funnel: in the baseline configuration only 116 of 415 settled frames (28%) have a perception lane and only 79 (19%) a planner path.

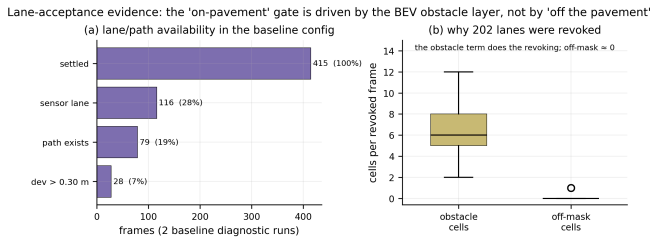


Fig. 25. Lane-acceptance evidence: baseline availability funnel (a) and the evidence behind the revoked lanes (b).

Run	Tick (ms)	Sensor read (ms)	Renderer frame (ms)	Sub-steps per frame	Effective command rate
Baseline acceptance (no path)	432	401	20	0	~2.2 Hz
Lane-gate factor arm	377	355	20	1 (max 5)	~5 Hz
Path-re-centring arm	403	~370	20	12	~31 Hz

A defect in the safety reading rule. The safety item used to judge every driving factor is a frame *count* of body-centre crossings, and a factor arm can travel twice as far as its control. 26 quantifies the consequence: the same arm can halve its per-metre crossing rate while showing a larger raw count. This is why, after the first two verdicts, we re-registered a per-distance reading (crossings per 100 m) alongside the raw maximum; the earlier verdicts stand as given, and we flag the rule defect as a methodological result rather than editing them.

implemented and run; the fifth was refuted by prior measurement before implementation.

F. Real-time behaviour and a metric defect

Control cadence. Per-frame telemetry shows the perception-and-planning tick costs 377–432 ms, of which 355 ms is sensor reading (camera ring 139–162 ms, range image 216–239 ms) and 11–13 ms is planning; the neural heads cost 3–5 ms. The renderer itself runs at ~50 fps (20 ms per frame), so the simulation is not the bottleneck: the control loop is paced at 2 Hz of wall clock by design and is in any case tick-bound at ~2.5 Hz. Between ticks a sub-step path re-runs the controllers, up to 15 sub-step iterations per tick, but it is gated on a path having been chosen: in the stall regime (no path) no sub-steps run and commands are issued every ~0.5 s, while in an arm where paths exist 12 sub-steps per frame yield about 31 command emissions per second. These are command-emission rates on the wall clock, not a claim about control bandwidth or about the age of the sensor data each command is computed from. Table IV summarises. The practical consequence is that the perceived "stutter" of the vehicle is a control-cadence artefact of the path-availability blocker, not a rendering or compute failure.

Table IV. Control cadence per configuration: tick cost, sub-step count and command-emission rate (wall clock).

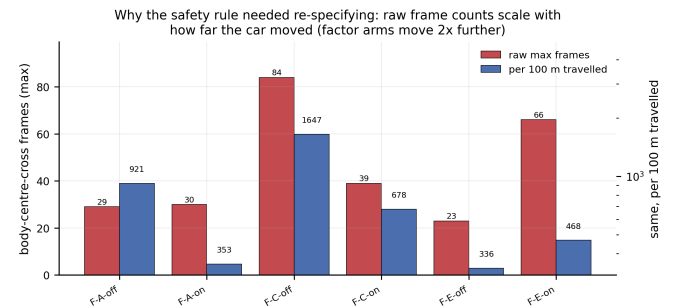


Fig. 26. Raw crossing counts versus the same quantity normalised per 100 m travelled (log scale).

VI. Discussion

What generalises. Five artefacts from this programme seem reusable beyond the specific stack. (i) The **provenance ladder with enforced credentials**: naming the truth rung of every label, and refusing to let a

command-line string promote one, made "which truth is this number measured against" an answerable question. (ii) The **counting contract**: reporting every ratio with its denominator, and keeping frame-level and instance-level counters separate, removed an entire class of "good number, empty pool" errors and made per-scene failures visible instead of averaged away. (iii) **Protocol versioning with a hash bound to the sealed set**: the mechanism is cheap and it caught a real mislabelling defect; without it, an evaluation produced under one definition would have been attributed to another. (iv) **Appearance certification of generated labels**: requiring that a claimed line be recoverable from the image turned "annotation noise" from a tuning problem into an admission decision. (v) **Pre-registered single-factor A/Bs with a fixed reading rule**, including an explicit refusal to treat unmeasured quantities as passes; this is what allowed four negative results to be reported as results rather than as failures.

What did not work. The scope gap is not closable by engineering alone, and it is worth being precise about why. About 30% of annotated line pixels fail the appearance predicate the adopted definition uses; the predicate reads RGB brightness and chroma only and computes no spatial contrast against the surrounding pavement, so what the measurement establishes is that those pixels do not satisfy *this* predicate. Whether they are visually identifiable to a human, recoverable from context or from another modality, or learnable at all are open questions that this work does not answer — the strong form of the claim ("invisible, therefore unlearnable") is not supported by the measurement. What is supported is the arithmetic: the paint scope retains 69% of annotated line pixels on the development pool, and the retained pixels are recalled at a higher rate than the removed ones (0.813 versus 0.736 at the same checkpoints), so conditioning the recall on the predicate is not a free improvement and is always reported alongside the full-label reading. Training on the removed pixels remains untested rather than refuted.

The offline gates do not transfer to closed-loop acceptance automatically either: the same perception that clears every offline gate leaves the car unable to move, for reasons (a static body-sweep deadlock) that no offline metric expresses. Finally, availability and safety trade against each other along the definition axis in a way that a single scalar cannot capture: in the four runs we have, the adopted definition was accompanied by no observed centre crossing and by a 66% relative drop in lane-source availability.

Implications for evaluation practice. Three practices would have changed our results materially. Report the *definition* alongside the metric, and make the definition a versioned artefact. Report per-distance or per-opportunity normalisations for safety counts, because motion confounds raw counts. And run one independent, one-shot confirmation whose composition differs from the development pool; our confirmation's negative result is the single most informative measurement in the programme.

VII. Limitations and Threats to Validity

No instance-level recall is claimed. The counting contract reports coverage, identity and role as conditional ratios over evaluated candidates; a missed marking on a frame where the perception produced no candidate enters pixel-level recall only. Readers who need instance recall should read the contract's definition (Section III-C) rather than infer one from $|R|/|C|$.

A small human anchor. The human-revised anchor is small (75 training frames) and the human-revised evaluation packs cover 9 packages (117 frames), so scene diversity is limited; the sealed one-shot set is engine-certified rather than human-revised, so its numbers are not directly comparable to the human-revised acceptance numbers. A larger human-revised held-out set would strengthen the definitional argument and is left to future work. The programme deliberately added **no new manual**

annotation during the reported rounds, which bounds both the cost and the coverage.

Single simulator, single map link. All perception data come from one simulator and, for the driving results, one town link. The definitional gap and the deadlock mechanism are measured on that link; their magnitude elsewhere is unknown. The offline sets span several certified scenes but not several maps.

Six seeds, and unequal seed counts. Acceptance uses six seeds; composition arms are reported with their seed counts (base 1, 4×6 , 6×1 , 8.5×6), and the $6 \times$ dose point rests on the single seed that was delivered. The dose result is therefore the location of the highest observed reading, not a stable optimum or an effect size; the balanced-seed comparison that would settle it is unfinished work. Frame counts inside a pool are highly correlated, so the seeds are the unit of variation we report and we do not treat pixels as independent trials.

The held-out set is consumed. The one-shot confirmation was performed once, as designed, and the set may not be read again; any future claim requires a newly sealed set. The instance-level probes were not run on it and remain UNKNOWN.

Artefact retention. Per-epoch training snapshots and superseded run directories were removed in a storage pass; every verdict file, manifest, per-frame telemetry trace, sealed frame and delivered checkpoint is retained, so all results and figures in this paper are reproducible from the retained artefacts, but some intermediate training states are not.

Driving results are from a research stack. The closed-loop results describe our stack, not automated driving in general; the acceptance gate is a project-internal target, and its thresholds were fixed before the runs and never relaxed.

VIII. Conclusion

We set out to make lane perception for automated driving measurable on a small, explicitly credentialed truth base, and to make its evaluation honest about definitions. A credentialed provenance ladder with four certification gates for its engine rung kept the machine-generated increment trustworthy while the human anchor stayed small and unchanged; a counting contract gave every reported ratio an explicit denominator; and a versioned, hash-bound protocol made the definition of "line" an artefact rather than an assumption. The resulting measurements are informative precisely where they are uncomfortable: the two natural definitions of a line each fail a different gate on the same data, and the disagreement is a third of the annotated pixels; the adopted definition passes all offline gates and removes every centre crossing in closed loop, at half the availability; and the one-shot independent confirmation does not reproduce the development numbers, which we report as the headline of that test. The closed-loop acceptance gate still fails, and we localised the blocker to a measurable deadlock mechanism rather than a tuning shortfall. Every factor, including five negative ones, was pre-registered and judged by rules fixed in advance; we consider that discipline, and the recorded negative results it produced, the main methodological contribution of this work.

Reproducibility and Data Availability

All results are regenerated from recorded artefacts: per-run scorecards, per-frame telemetry traces, run manifests, gate verdict files, and sealed-set records. Two scripts regenerate every figure in this paper and the supplement from those files; the protocol hash, the sealed-set digest and the one-shot confirmation record are printed inside the figures so that a reader can verify which definition and which data produced them. Training pools, checkpoints, the sealed set and the audit of removed artefacts are

retained in the project repository; per-epoch snapshots removed during a storage pass are listed in a machine-readable audit file.

Nomenclature

Symbol	Meaning
P	frames carrying at least one line candidate
C	extracted candidate instances
C_{out}	candidates that fall on frames outside P
R	reference instances available from the annotation
M	matched instances, $M \subseteq R$
L	instances whose lateral placement agrees, $L \subseteq M$
A	instances whose left/right role agrees, $A \subseteq L$
coverage	
identity	
role	
precision, recall	pixel level, reported in label scope and in paint scope
Abbreviation	Meaning
FSD	full self-driving (perception-led driving stack)
BEV	bird's-eye view
IoU	intersection over union
TTC	time to collision
SWA	stochastic weight averaging
PLC	painted-line centring corrector
A/B	alternating control/factor comparison
UNKNOWN	measured quantity unavailable; never read as a pass

Open problems and unfinished work

The programme is deliberately reported with its unfinished parts in the same detail as its results; this section lists them with the evidence that localises each one and the concrete next experiment.

1. The closed-loop acceptance gate is unmet. Four runs of the acceptance configuration fail 0/4 on the benchmark's hard targets (stall fraction 0.93–1.00, travelled 0.9–14.5 m in 120 s, 76.5–88.8 m remaining to goal), while the measured safety items hold (0 collisions, 0 off-road frames, 0 reversing). The blocker is localised to a static deadlock: a stationary car, off-centre by 0.86 m, whose inflated body sweep crosses the lane boundary within a median 2.50 m, so the path the planner generates is refused and the car cannot re-centre (Section V-E, Figs. 23 and 24). The record shows that the generated path is refused, not that no feasible path exists; separating the two needs a controlled sweep over frozen poses that stores the perception boundary, the candidate paths, the sweep geometry and the final command for both executable and non-executable cases. *Next experiment:* the three factors that engaged (bounded hold window, path re-centring, lane gate) were each judged alone; the natural successor is a single pre-registered **combined** arm against the best single-factor arm, plus the plan-layer decisions that the programme may not take unilaterally — the planned-crossing hard threshold and the semantics of the on-pavement gate, both of which are safety rules.

2. Lateral-reference quality is the next perception-side lever. The planner's path sits a median 0.773 m from its own lane reference while the two layers' references agree exactly (0.0 m gap), and the bounded centring corrector engages only where line evidence exists (39 of 805 settled frames). *Next experiment:* raise line-evidence availability (a stronger line head, or a fusion that does not depend on the sparse line class), or move

the centring upstream so the path is planned on the lane reference rather than corrected after the fact.

3. Balanced-seed dose comparison. The dose response is the weakest of the reported composition results: the 6× point rests on one seed while 4× and 8.5× rest on six. The checkpoints are retained, so the missing work is evaluation, not training: score the same pools under one frozen protocol with paired seeds and equal counts, report per-seed differences and intervals, and treat the small-sample intervals as exploratory. If the peak does not survive, the delivered composition is still the arm that was selected before the confirmation, and the paper's claims about it are bounded accordingly.

4. Control cadence. The perception-and-planning tick costs 377–432 ms, of which 355 ms is sensor reading (camera ring plus range image) and 11–13 ms is planning; the control loop is paced at 2 Hz by design and is tick-bound at ~2.5 Hz. Sub-steps smooth the interval at up to 15 Hz but are gated on a path having been chosen, so the stall regime is exactly where they do not run. *Next experiment:* cheaper sensor reads (ring resolution or round-robin mounts, range reuse under the existing keep-alive) measured as a single factor with both the tick cost and the driving hard targets as read-outs.

5. Specification of safety metrics. The driving factors were judged with a frame-count maximum, and a factor arm can travel twice as far as its control; Fig. 26 quantifies how the same arm can halve its per-metre crossing rate while showing a larger raw count. A per-distance reading has been pre-registered for new rounds; the earlier verdicts stand as recorded, and the rule defect is reported as a methodological result rather than edited away.

6. The definitional gap is not closable by engineering. About 30% of annotated line pixels are dark paint with no local contrast; no local-contrast model can recover them, and training on them teaches hallucination (a documented negative result). *Options:* an explicit convention change (exclude them from the recall denominator, as the paint scope does), an additional modality, or a prior over road geometry — each of which changes the definition and must therefore be pre-registered as such.

7. Truth provenance and held-out capacity. The human-revised anchor is small (75 training frames; 9 evaluation packages, 117 frames), and the sealed one-shot set is engine-certified and now **consumed**: any further confirmation requires a newly sealed set. *Next experiment:* a larger human-revised anchor and a newly sealed human-revised held-out set, which would also let the confirmation's numbers be compared against the acceptance numbers on the same truth rung.

8. Scale and external validity. All results come from one simulator, the driving results from one town link, the acceptance from six seeds, and the composition arms from one to six seeds (the 6× dose point rests on a single seed). No real-vehicle data were used. The dose conclusions are therefore about a peak location, not an effect size, and the definitional findings should be re-measured on a second map before being treated as general.

9. Instance vocabulary for harder geometry. The counting contract's role vocabulary (left/right, near/far) is defined for lane markings on a single carriageway; junctions, 3-D lane estimation, and multi-lane role conventions are undefined, and the identity gate's failure mode on the baseline definition is exactly a role/instance ambiguity rather than a pixel error.

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Author contributions. The author is the sole author and performed every role: conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualisation, supervision.

Ethics. Not applicable: the work uses a commercial driving simulator and no human subjects. The human annotation referenced in this paper was performed by the author on his own simulator recordings.

Use of AI tools. AI coding assistants (DeepSeek and OpenCode) were used as development tools during the implementation and documentation of this work; all experimental design, measurements, verdicts and conclusions were produced and verified by the author, and every quantitative claim in this paper is regenerated from recorded artefacts by the scripts listed below.

Data and code availability. The project repository is <https://github.com/Qiongkura/BeamNG-autopilot> (branch `fix/round3-hardening-20260921`). All quantitative claims are regenerated from recorded artefacts retained in that repository: per-run scorecards, per-frame telemetry traces, run manifests, gate verdict files, sealed-set records and the protocol hash. Three scripts regenerate every figure in this paper and its supplement from those files (`m5_paper_figures.py`, `m5_paper_figures_ext.py`, `m5_paper_figures_extra.py`); `build_paper.py` regenerates this manuscript's LaTeX and HTML from a single Markdown source. Per-epoch training snapshots and superseded run directories removed in a storage pass are listed in a machine-readable audit file.

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Appendix A: Recorded artefacts behind each results section

Results section	Artefact(s)
V-A acceptance	r2_verdict_v7_20261005. json, r2_verdict_v8_ADOPTED_20261005. json
V-B definitional gap	r2_verdict_v8_ADOPTED_20261005. json (per-seed label and paint recall)
V-B boundary map	r2_verdict_v8_paint_{nodilate_mean, dil1b, dil2}_20261005. json
V-B appearance gate	dev_pixel_{base, appgate}_20261004. json, r3b_pixel_{lat3, appgate}_20261004. json
V-B scans	dev_scan_{lateral, parallel}_20261004. json, r3b_scan_{lateral, parallel}_20261004. json
V-B identity scopes	t16_dual_scope_10seed. json

Results section	Artefact(s)
V-C dose response	dev_full_arms_20261005. json, dev_full_dose_arms_s4{3..7}. js
V-C rejected factors	r2_verdict_swa_20261005. json, dev9_beta08_v8d1_20261005. js
V-D confirmation	final_set_v5_20261005/seal/{final_set_seal. json, confirmati seed42. json, final_set_ledger. jsonl}
V-D composition audit	final_pool_audit_20261005. json, collect_isolation_20260927.
V-E driving A/B)	logs/fsd_benchmark/{scorecard, manifest, town}*_*. json (all runs
V-F timing	timing_retest_20260927. json, per-frame telemetry frame_ms/tick_
IV pool screening	t16_r3_density_dev. json, t16_r3_pool_{,b}_density_scenes. js
IV cost ledger	el_negative_pool_20260927. json gpu_ledger_machine. json

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